LEARNER GUIDE

Reverse percentages

Percentages, Ratio, Growth and Bounds · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **reverse percentage** problem gives you the value **after** a change and asks for the **original value**. The percentage is always a percentage of the original, so after = original × multiplier. To undo multiplying, divide the after value by the multiplier. Taking the percentage of the after value instead is a common slip: it uses the wrong amount.

What you are learning to notice

- First decide: is the value you have the before value or the after value?
- The percentage belongs to the original, never to the after value.
- To undo × 1.2, divide by 1.2. To undo × 0.85, divide by 0.85.
- Always check forwards: original × multiplier should give the after value.
- A 100% decrease cannot be undone: the multiplier is 0.

Moving through it, one step at a time

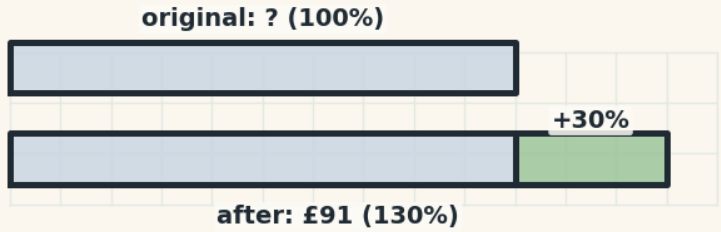
Write the multiplier for the change (a 20% rise is 1.2; 15% off is 0.85). Divide the after value by the multiplier. Then check forwards by multiplying your answer by the multiplier.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Undoing an increase

After a 30% rise, a bike costs £91. What was the original price?

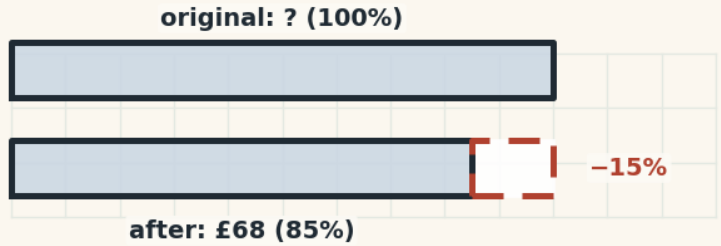


1. Which value do you have?	£91 is the value after the rise
2. The multiplier	100% + 30% = 130% = 1.3
3. Undo by dividing	$\frac{91}{1.3} = 70$
4. Check forwards	£70 × 1.3 = £91

The original price was £70 (not £91 – 30% of £91 = £63.70).

Example 2 — Undoing a decrease

After 15% off in a sale, a coat costs £68. What was the original price?



1. The multiplier	100% – 15% = 85% = 0.85
2. Undo by dividing	$\frac{68}{0.85} = 80$
3. Check forwards	£80 × 0.85 = £68

The original price was £80 (not £68 + 15% of £68 = £78.20).

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (percentages as multipliers (Lesson 3) and percentage change (Lesson 4)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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